Collège Lycée Suger **Projets Scientifiques**

Année 2025-2026 **Classes de 6ème-5ème**

Suger Carbon Footprint – Google Sheet activity

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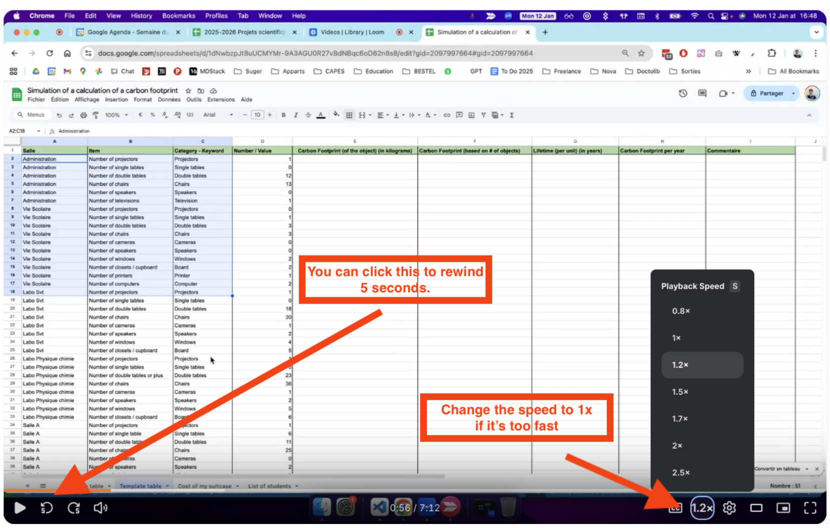
1 Access to / Description of the Google Sheet

How to read this document
- Red cells in the instructions: you must write a value or a formula in these cells.
- **Green boxes** (Indication): help, explanations, and common mistakes.

1. Watch the video at the following link: Simulation of a calculation of a carbon footprint. See how to interact the video in the image below :



How to interact with the Loom video

2. Make sure you can click on a tab that has your name on it.
   - If you cannot find it, please send me an email in EcoleDirecte.
   - Do **not** edit the **Template Table** nor the **Reference Table** tab.

Indication - Description of your tab

- The main table shows the inventory made by students in a subset of rooms in Suger (tables, projectors, etc.).
- Definitions of the main columns:
  - Column A (Salle): the room where the inventory was made.
  - Column B (Item): what was counted in that room.
  - Column C (Category - Keyword): a simple word to designate what was counted.
  - Column D (Number / Value): the quantity counted in that room, for that specific item.
- To better understand the other columns, watch this video: [Suger Carbon Footprint] Description of the main Google Sheet tabs.

2 Inventory: Filling your own Google Sheet tab

Indication – Before you start

- You will write formulas in Google Sheets. If you need a reminder, watch: [Suger Carbon Footprint] Using formulas in Google Sheet.
- Work only in **your own tab**. Do not touch the **Template**.

2.1 Total Carbon Footprint of each object

Objective: fill the column ***Carbon Footprint (of the object) (in kilograms)*** *using a reference table, then compute the total based on the number of objects.*

1. In your own Google tab, in cell **E2**, build a formula using the **VLOOKUP** function to fill the column **Carbon Footprint (of the object) (in kilograms)**.
   - You should watch this tutorial to better understand how to use VLOOKUP: [Suger Carbon Footprint] How to use the VLOOKUP formula.
   - The value must be returned automatically based on the **Category - Keyword** column.
   - Use the **Reference table** tab to get the right values.

   Indication – Using VLOOKUP

   - If you use ChatGPT for help, include:
     - You are using **Google Sheets**.
     - You want to use **VLOOKUP**.
     - The reference table is in another tab named **Reference table**.
     - You want to return a value from a specific column, based on a matching keyword.
   - Expected value in cell E2: **900**.

2. Once you have filled cell E2 with the formula, use autocomplete down to cell E64 to propagate the formula.
   - If you see an error, copy/paste the error message and the formula to ChatGPT to debug it.

   Indication – Common error with VLOOKUP

   - **#N/A** often means: the keyword in **Category - Keyword** does not exist in the **Reference table**.
   - Another common reason: your lookup range moves when you propagate the formula.
   - Fix: use absolute references for the reference table range (with **$**).
   - Example: use **'Reference table'!$A$1:$F$13** instead of **'Reference table'!A1:F13**.

   - At the end of question 2, all column E should contain numerical values (no error).

3. In cell **F2**, write a formula which calculates the **Carbon Footprint (based on # of objects)** based on:
   - the number of objects in column D (**Number / Value**)
   - the value in column E (**Carbon Footprint (of the object) (in kilograms)**)
4. Now propagate the formula from cell F2 down to cell F64 so that all column F is filled with numerical values.

2.2 Lifetime coefficient applied to the Carbon Footprint

Context: the carbon footprint of constructing a projector may be 900 kg of CO_2. If it is used for 5 years, the annual footprint is $900/5 = 180$ *kg of CO_2 per year.*
Objective: fill the object lifetime, then compute the annual footprint per line.

1. In cell **G2**, using the VLOOKUP function, build a formula which determines the **Lifetime (per unit) (in years)** of the object by looking up its value in column E of the **Reference table** tab.
   - Expected value in G2: **5**.
2. Now propagate the formula from cell G2 down to cell G64 so that all column G is filled with numerical values.
3. In cell **H2**, build a formula which calculates the **Carbon Footprint per year** by dividing:
   - the value in column F (**Carbon Footprint (based on # of objects)**)
   - by the value in column G (**Lifetime (per unit) (in years)**)
4. Propagate this formula down to cell H64.

2.3 Total Annual Carbon Footprint of Suger inventory

Context: you now have an annual carbon footprint for each line of the inventory. The next step is to compute the total annual carbon footprint of all items in Suger.

1. In cell **K1**, build a formula which calculates the **Total Annual Carbon Footprint of Suger objects** by summing all the values in column H (**Carbon Footprint per year**).

3 iPad carbon footprint

Objective: estimate the annual carbon footprint of iPads in Suger, using a manufacturing footprint and a lifetime hypothesis.

1. Using the https://impactco2.fr/ website, determine the approximate carbon footprint of an iPad in kilograms of CO_2 (focus only on manufacturing).
   - Write your answer in cell **K5**.
   - Copy/paste the URL of the page where you found the information in cell **L5**.
2. Using the internet or ChatGPT, determine the average lifetime of an iPad in years.
   - Write your answer in cell **K6**.
   - Copy/paste the URL of the page where you found the information in cell **L6**.
3. In cell **K7**, build a formula which calculates the **Annual Carbon Footprint of an iPad** by dividing:
   - cell K5 (Carbon Footprint of an iPad)
   - by cell K6 (Average lifetime of an iPad in years)
4. In cell **K8**, put an estimated value of the number of students in Suger (you can check on the Internet, on the Suger website, or elsewhere).
   - Copy/paste your source in cell **L8**.
5. In cell **K9**, build a formula which calculates the **Total Annual Carbon Footprint of iPads in Suger**, based on the available information.

4 Conclusion

Congrats. You've learned how to use Google Sheets to compute the carbon footprint of Suger school equipment. It will now be fairly easy to apply what you've done on the actual document where the full inventory was made. We're almost there !